

Quick Check

Lesson 6: Keeping Cool and Staying Warm: Thermoregulation in the Dog, the Lizard and Us

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.3: Excretion and Homeostasis in Animals | Lesson 6

On a hot afternoon in Kericho, a dog pants rapidly with its tongue hanging out. How does panting help the dog control its body temperature?

- A** The dog sweats through its tongue, just as humans sweat through the skin
- B** Panting warms the dog's blood so it matches the air temperature
- C** Panting makes the dog's body produce more heat so it can adapt to the weather
- D** Water evaporates from the tongue and airways and carries heat away from the body

Blood in a mammal becomes too hot. Which response in the skin helps to lower body temperature?

- A** Vasodilation, which brings more blood close to the skin surface so heat is lost
- B** Vasoconstriction, which reduces blood flow near the skin surface
- C** Shivering, which makes the muscles contract rapidly
- D** Hair erection, which traps a layer of air next to the skin

A field note says: 'Agama lizard on a rock at 9 am and 11 am. At 2 pm, when the rock is very hot, it moves into the shade.' What is the best explanation for this behaviour?

- A** The lizard is lazy and rests when it is tired
- B** The lizard produces its own heat and moves to the shade to burn it off
- C** The lizard cannot control its temperature internally, so it moves to avoid overheating
- D** The lizard's hypothalamus makes it sweat in the shade

Body temperature control is called a negative feedback loop. Why is it called 'negative'?

- A** The response makes the body temperature change even more in the same direction
- B** Negative feedback is harmful to the body and must be avoided
- C** The response reverses the change that triggered it and brings the temperature back to normal
- D** The control centre switches off and stops responding to temperature changes

A goat is standing in the cold early morning and its body temperature begins to fall. Which set of responses helps it to conserve and produce heat?

A Vasoconstriction, shivering and hair erection

B Vasoconstriction, panting and hair erection

C Vasodilation, sweating and panting

D Vasodilation, shivering and sweating

Some frogs and lungfish bury themselves in mud during the long dry season at the shores of Lake Victoria and become inactive. What problem does this response (aestivation) help them survive?

- A** Too much water in their bodies
- B** Prolonged heat and drought
- C** Prolonged cold weather
- D** A lack of food caused by too many predators

A dog pants for a long time on a hot day. Besides cooling the dog, what is another effect of the panting?

- A** The dog's body stops using the hypothalamus because it is already cooling
- B** The dog's kidneys make more dilute urine to remove the extra water
- C** The dog loses water by evaporation, so its body must conserve water and the dog needs to drink
- D** The dog gains water from the air it breathes in

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).